Injector to Governor Linkages

The output of the unit injectors is governor controlled by means of a set of linkage. Governor control rods extend from the governor housing and connect to a control tube lever by means of a clevis and pin: linear movement of the governor control rods will by this means be converted into rotary movement of the control tube which runs lengthwise through the upper cylinder head. Extending from the control tube are the rack levers.
The rack levers link the individual unit injectors with the control tube; when the control tube is rotated by the governor control rods, the rack levers extending from the control tube will linearly move the unit injector racks. A critical procedure in DDC engine tune-up is ensuring that the unit injectors are balanced, that is, that the point of register of the helices with the bushing ports in each unit injector in the engine is identical. This is achieved by correctly setting the control tube adjusting screw; this adjusting screw locates the radial position of the rack lever.

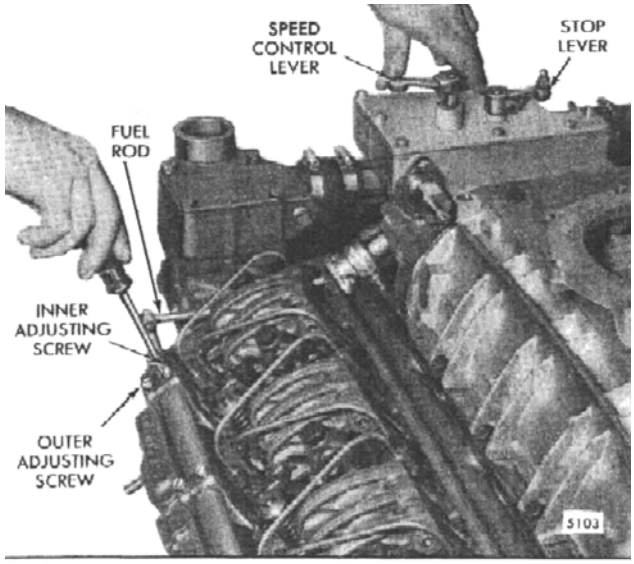


Fig. 8 – Positioning No. 1L Injector Rack Control Lever (Two Screw Assembly)

Governor

A simple mechanical governor is used in most truck and bus applications. Governor components consist of a set of engine driven flyweights, a thrust collar and spring set. As in most mechanical governors, centrifuge exacted by the flyweights attempts to diminish engine fueling while spring force moderated by speed control lever position attempts to increase engine fuelling. The thrust collar acts as an intermediary. Both limiting speed and variable speed governors are used: the governor type should be identified prior to engine tune-up as the procedure varies with governor type.

ENGINE TUNE-UP PROCEDURES

There is no scheduled interval for performing an engine tune-up. As long as the engine performance is satisfactory, no tune-up should be needed. Minor adjustments in the valve and injector operating mechanism, governor, etc should only be required periodically to compensate for normal wear on parts.

T o comply with emissions regulations, injector timing, exhaust valve clearance, redline idle and no-load speeds, and throttle delay or fuel modulator settings must be checked and adjusted if necessary, at 50,000 miles intervals.

The type of governor used depends upon the engine application. Since each governor has different characteristics, the tune-up procedure varies accordingly. The following types of governors are used.

1. Limiting speed mechanical.

2. Variable speed mechanical

3. Hydraulic

The mechanical governors are identified by a name plate attached to the governor housing. The letters D.W. L.S. stamped on the name plate denote a double weight limiting speed governor. A single-weight variable speed governor name plate is stamped S.W.-V.S.
The engine we will be working on is a D.W.-L.S. so we will concentrate on the procedure for this type of governor but note that the procedures will change slightly for each type so always consult the appropriate service manual.
Normally when performing a tune-up on an engine in service. it is only necessary to check the various adjustments for a possible change in the settings However, if a cylinder head, governor or injectors have been replaced or overhauled, then certain tune-up adjustments are required. Accurate tune-up adjustments are very important if maximum performance and economy are to be obtained.

NOTICE: if a Supplementary governing device, such as the throttle delay mechanism is used, it must be disconnected prior to the tune-up. After the governor and injector rack adjustments are completed the supplementary governing device must be reconnected and adjusted.